

Name: subhash

Assignment 3: Auto ML with akkio or jadbio or any other good zero code automl

Get akkio.com freeaccount, demonstrate one (or more if you are interested) end2end ml project in jadbio (no code project).

I have created a free Akkio account and developed a Customer Churn Prediction machine learning model using it. I have uploaded the video on YouTube, demonstrating the end-to-end process.

Customer Churn Prediction: This project helps us predict which customers will continue doing business and which ones will not.

1. **Dataset:** I used a Kaggle dataset for this project, which contains columns such as customerID, gender, SeniorCitizen, Partner, churn, and more.

Kaggle Dataset Link: Customer Churn Prediction

Dataset Link:

<https://www.kaggle.com/code/bhartiprasad17/customer-churn-prediction/notebook>

2. The model automatically split the dataset into testing and training sets, performed feature training, and selected the best algorithm for the task. The model's performance was evaluated using metrics like accuracy, precision, and recall.

Project Objective

The goal of this project is to predict which customers are likely to churn (stop using a service) based on various demographic and behavioral features. Accurate predictions can help businesses proactively retain customers and optimize their marketing efforts.

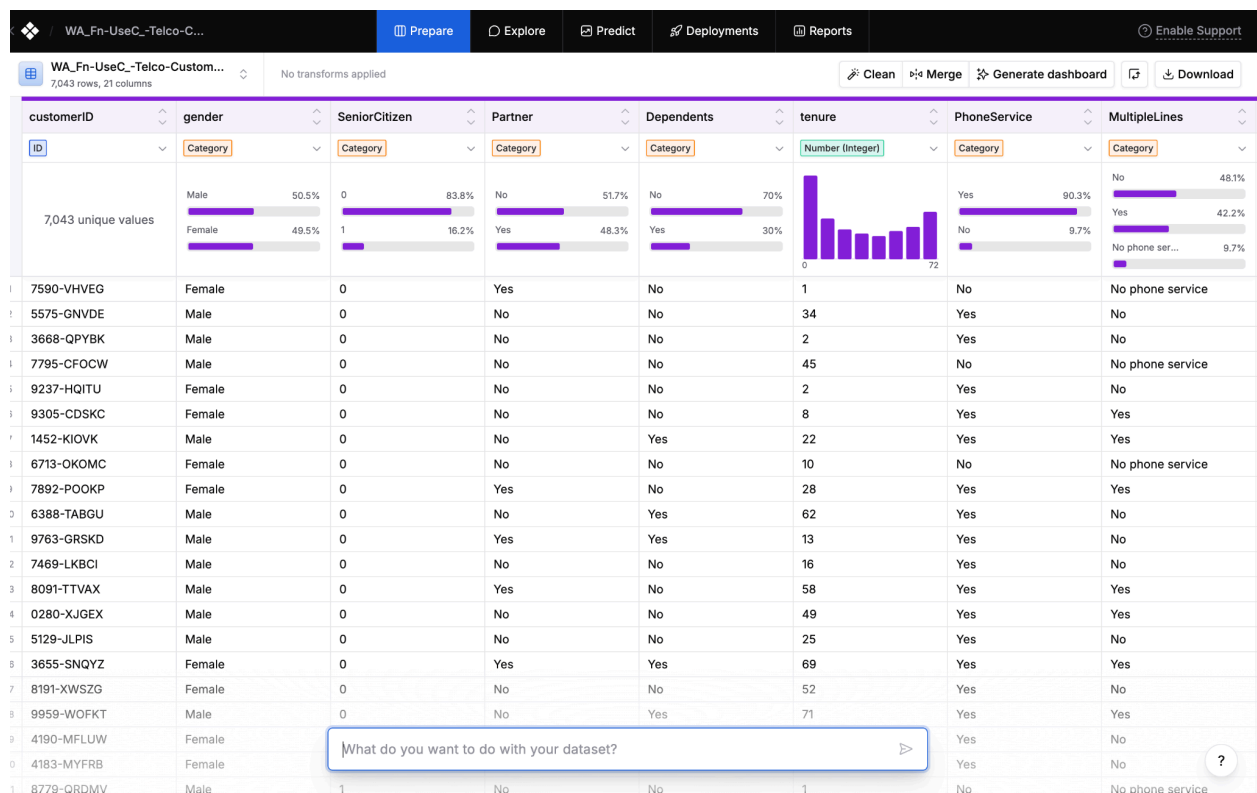
Step-1: Prepare

1. Dataset Preparation

I selected a publicly available dataset from Kaggle titled **Customer Churn Prediction**. The dataset includes features like:

- customerID
- gender
- SeniorCitizen
- Partner
- tenure
- MonthlyCharges
- Churn (target variable)

The dataset required minimal preprocessing, as Akkio's platform handles much of the data cleaning and preparation automatically.

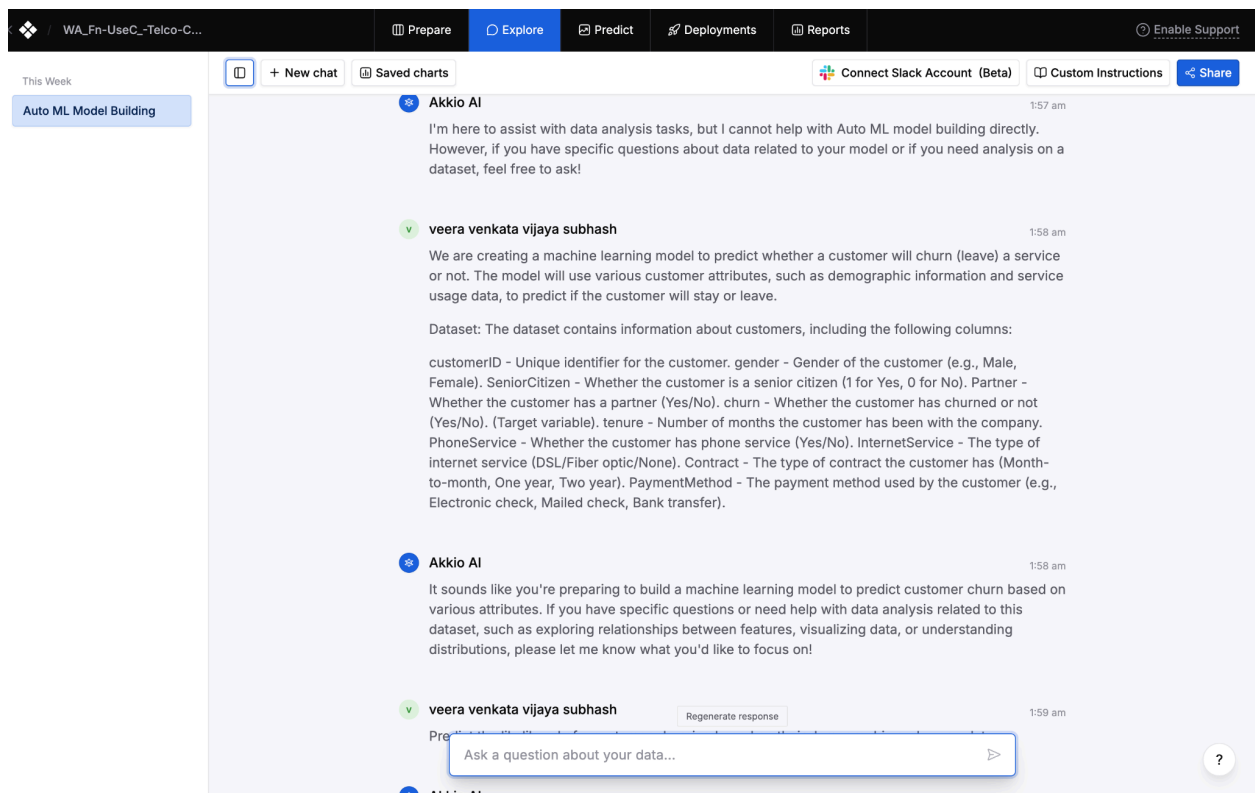


Step-2: Explore

Exploring the Dataset

Akkio's intuitive interface made it easy to analyze the data visually. Key steps during this phase included:

- Understanding feature distributions and their relationships with the target variable (**Churn**).
- Identifying trends in customer demographics that influence churn (e.g., senior citizens and high monthly charges).



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Electronic check, mailed check, bank transfer).

Akkio AI

1:58 am

It sounds like you're preparing to build a machine learning model to predict customer churn based on various attributes. If you have specific questions or need help with data analysis related to this dataset, such as exploring relationships between features, visualizing data, or understanding distributions, please let me know what you'd like to focus on!

veera venkata vijaya subhash

1:59 am

Predict the likelihood of a customer churning based on their demographic and usage data.

Akkio AI

1:59 am

To predict the likelihood of a customer churning based on their demographic and usage data, we first need to analyze the dataset to understand the relationships between the features and the target variable (churn). This involves several steps, typically including:

1. Exploratory Data Analysis (EDA): Understand the distribution of features and their relationship with churn.

2. Data Preprocessing: Handle missing values, convert categorical variables into numerical format, and scale features if necessary.

3. Model Selection: Choose appropriate algorithms for prediction (e.g., logistic regression, decision trees, random forests, etc.).

4. Model Training and Evaluation: Train the model on a training dataset and evaluate its performance on a test dataset.

Would you like me to perform an exploratory data analysis (EDA) to help you understand the relationships in your dataset, or do you have a specific aspect of the analysis in mind?

Regenerate response

veera venkata vijaya subhash

1:59 am

Ask a question about your data...

?

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Customers who churn: 274, 444

Customers who do not churn: \$61.27

Higher monthly charges are associated with a higher likelihood of churn, indicating that pricing may influence customer retention.

6. Customer Count by Gender and Churn

Total customers:

Male (no churn): 2549, (churn): 939

Female (no churn): 2625, (churn): 930

The number of male and female customers who churn is quite balanced, suggesting that gender may not be a strong predictor of churn in this dataset.

7. Customer Count by Internet Service and Churn

Internet Service Type:

Customers with DSL: 1962 (no churn), 459 (churn)

Customers with Fiber optic: 1799 (no churn), 1297 (churn)

Customers with No service: 1413 (no churn), 113 (churn)

Customers with fiber optic service have a higher churn rate compared to those with DSL, indicating that service type may play a role in customer satisfaction and retention.

Conclusion

These insights provide a strong foundation for building a predictive model for churn. Key features to consider include tenure, contract type, average monthly charges, and the presence of senior citizens. Addressing potential class imbalance in the dataset will be crucial for accurate predictions.

If you need further analysis or assistance with model development, please let me know!

Regenerate response

veera venkata vijaya subhash

2:00 am

Ask a question about your data...

?

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consider include tenure, contract type, average monthly charges, and the presence of senior citizens. Addressing potential class imbalance in the dataset will be crucial for accurate predictions.

If you need further analysis or assistance with model development, please let me know!

veera venkata vijaya subhash

2:00 am

Please select a 'Predict' field to create a predictive model

Akkio AI

2:00 am

In the context of predicting customer churn, the most appropriate field to use as the target variable (or "Predict" field) is the **"Churn"** column. This column indicates whether a customer has churned (Yes) or not (No), making it the ideal target for your predictive model.

Target Variable:

- Churn: This field will be used to predict the likelihood of a customer leaving the service.

Features:

You can use various features from the dataset as predictors, including:

- customerID: Unique identifier (not typically used in modeling).
- gender: Categorical variable indicating the gender of the customer.
- SeniorCitizen: Indicates if the customer is a senior citizen (1 for Yes, 0 for No).
- Partner: Indicates if the customer has a partner (Yes/No).
- tenure: Number of months the customer has been with the company.
- PhoneService: Indicates if the customer has phone service (Yes/No).
- InternetService: Type of internet service (DSL/Fiber optic/None).
- Contract: Type of contract (Month-to-month, One year, Two year).
- PaymentMethod: Payment method used by the customer.

Ask a question about your data...

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Based on the analysis of the dataset, here are some key insights regarding customer churn and its influencing factors:

- Churn Distribution
 - Proportion of Churn:
 - No Churn: 73.5%
 - Churn: 26.5%

This indicates that a significant majority of customers do not churn, which is important for model training as it may lead to class imbalance.
- Average Tenure by Churn
 - Customers who churn have an average tenure of approximately 18 months.
 - Customers who do not churn have a substantially higher average tenure of about 38 months.

This suggests that longer-tenured customers are less likely to churn, which could be a critical feature for prediction.
- Senior Citizen Percentage by Churn
 - Senior Citizens:
 - Percentage of Senior Citizens among those who churn: 25.5%
 - Percentage of Senior Citizens among those who do not churn: 12.9%

This shows that senior citizens are more likely to churn compared to younger customers.
- Churn Rate by Contract Type
 - Churn Rates based on contract types:
 - Month-to-month: 42.7% churn
 - One year: 31.9% churn
 - Two year: 11.9% churn

Ask a question about your data...

This highlights that customers on longer-term contracts (one year and two years) are significantly

Step:3 Predict:

Predictive Model Creation

Using Akkio, the predictive modeling process was streamlined:

- The dataset was automatically split into training and testing sets.
- Akkio's AutoML engine selected the optimal machine learning algorithm for the task.
- Performance metrics such as **accuracy**, **precision**, and **recall** were used to evaluate the model. These metrics indicated the model's ability to distinguish between churned and non-churned customers effectively.

The screenshot displays the Akkio AI web interface. The top navigation bar includes tabs for 'Prepare', 'Explore' (active), 'Predict', 'Deployments', 'Reports', and 'Enable Support'. Below the navigation bar, there are buttons for '+ New chat', 'Saved charts', 'Connect Slack Account (Beta)', 'Custom Instructions', and 'Share'. The main content area is titled 'Akkio AI' and shows a chat log with the following insights:

Based on the analysis of the dataset, here are some key insights regarding customer churn and its influencing factors:

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 - **Proportion of Churn:**
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- Average Tenure by Churn**
 - Customers who churn have an average tenure of approximately 18 months.
 - Customers who do not churn have a substantially higher average tenure of about 38 months.

This suggests that longer-tenured customers are less likely to churn, which could be a critical feature for prediction.
- Senior Citizen Percentage by Churn**
 - **Senior Citizens:**
 - Percentage of Senior Citizens among those who churn: 25.5%
 - Percentage of Senior Citizens among those who do not churn: 12.9%

This shows that senior citizens are more likely to churn compared to younger customers.
- Churn Rate by Contract Type**
 - **Churn Rates based on contract types:**
 - Month-to-month: 42.7% churn
 - One year: 11.9% churn

This highlights that customers on longer-term contracts (one year and two years) are significantly less likely to churn.

At the bottom of the chat log, there is a text input field with the placeholder 'Ask a question about your data...' and a 'Regenerate response' button.

WA_Fn-UseC_-Telco-C...

PrepareExplorePredictDeploymentsReports

Enable Support

This Week

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Share

Akkio AI

1:59 am

Based on the analysis of the dataset, here are some key insights regarding customer churn and its influencing factors:

1. Churn Distribution

• Proportion of Churn:

No Churn: 73.5%

Churn: 26.5%

This indicates that a significant majority of customers do not churn, which is important for model training as it may lead to class imbalance.

2. Average Tenure by Churn

• Customers who churn have an average tenure of approximately 18 months.

• Customers who do not churn have a substantially higher average tenure of about 38 months.

This suggests that longer-tenured customers are less likely to churn, which could be a critical feature for prediction.

3. Senior Citizen Percentage by Churn

• Senior Citizens:

Percentage of Senior Citizens among those who churn: 25.5%

Percentage of Senior Citizens among those who do not churn: 12.9%

This shows that senior citizens are more likely to churn compared to younger customers.

4. Churn Rate by Contract Type

• Churn Rates based on contract types:

Month-to-month: 42.7% chu

Generate response

Generate 11.9% churn

Ask a question about your data...

?

This highlights that customers on longer-term contracts (one year and two years) are significantly

WA_Fn-UseC_-Telco-C...

PrepareExplorePredictDeploymentsReports

Enable Support

WA_Fn-UseC_-Telco-Customer-C...

No transforms applied

Retrain Model

Back to Training Types

Predict

Predict Fields

Select which numerical or categorical fields to predict and optionally ignore.

Predict

Ignore

Search fields...

gender

SeniorCitizen

Partner

Dependents

tenure

PhoneService

MultipleLines

InternetService

OnlineSecurity

Training Mode

Choose how long you would like to train your model. Longer training generally improves model accuracy.

Fastest

Advanced Settings

Prediction Column

Contract CATEGORY

Outcome of Interest

Two year Edit

Expand Insights

Segments

Sets of similar records in your dataset grouped by outcome of interest

Likely Two year

Long-Term Contract

This segment has the longest tenure compared to the other segments and most customers have a two-year contract with the company, with bank transfer as their preferred payment method.

Two year

In this segment

2.6x

vs base rate

Segment size

34.8%

of all records

Uncertain

Average Tenure

This segment has an average tenure and most customers in this segment use mailed check as their payment method.

Two year

In this segment

0.3x

vs base rate

Segment size

39.8%

of all records

Unlikely Two year

Short-Term Contract

This segment has the shortest tenure compared to the other segments and most customers have a month-to-month contract with the company, with electronic check as their preferred payment method. These customers are also more likely to churn.

Two year

In this segment

0.0x

vs base rate

Segment size

25.4%

of all records

Key Similarities

TechSupport

is

Yes

49.2%

Segment Rate / Segment Average

28.2%

Base Rate / Base Average

1.7x

Rate vs. Baseline / Average Delta

OnlineSecurity

is

Yes

46.9%

Segment Rate / Segment Average

28.2%

Base Rate / Base Average

1.7x

Rate vs. Baseline / Average Delta

DeviceProtection

is

No internet service

36.9%

Segment Rate / Segment Average

22.2%

Base Rate / Base Average

1.7x

Rate vs. Baseline / Average Delta

StreamingMovies

is

No internet service

36.9%

Segment Rate / Segment Average

22.2%

Base Rate / Base Average

1.7x

Rate vs. Baseline / Average Delta

StreamingTV

is

No internet service

36.9%

Segment Rate / Segment Average

22.2%

Base Rate / Base Average

1.7x

Rate vs. Baseline / Average Delta

+ See more

Step 4: Deployment

Deployment

Once the model was trained, Akkio provided straightforward deployment options:

- The model could be integrated into external tools or used to generate batch predictions for future datasets.
- This feature is particularly valuable for business applications, where real-time insights are essential.

The screenshot displays the Akkio web interface for deploying a model. The top navigation bar includes tabs for Prepare, Explore, Predict, Deployments (active), and Reports. The main content area is divided into two sections: 'Select Fields' on the left and 'Sample request/response' on the right.

Select Fields: A list of features is shown with checkboxes. The 'Select All' option is checked. The following fields are selected:

- customerID
- gender
- SeniorCitizen
- Partner
- Dependents
- tenure
- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- PaperlessBilling
- PaymentMethod
- MonthlyCharges
- TotalCharges
- Churn

Sample request: A curl command is provided for a POST request to the Akkio API. The request body is a JSON object containing the selected fields.

```
curl -G https://api.akkio.com/api \
-d project_key="a8edf55a-1b3c-4fa2-8324-9bd016988df3/1" \
-d api_key="3150d52e-83a0-46c4-b594-4eb936fd66ff" \
--data-urlencode \
data='[{"SeniorCitizen": "0", "Dependents": "No", "tenure": "1", "InternetService": "DSL", "OnlineSecurity": "No", "DeviceProtection": "No", "TechSupport": "No", "StreamingTV": "No", "Stream check", "Churn": "No"}]' \
-d deploy-transforms-only=false'
```

Sample response: A JSON object is shown, representing the predicted values for the selected fields.

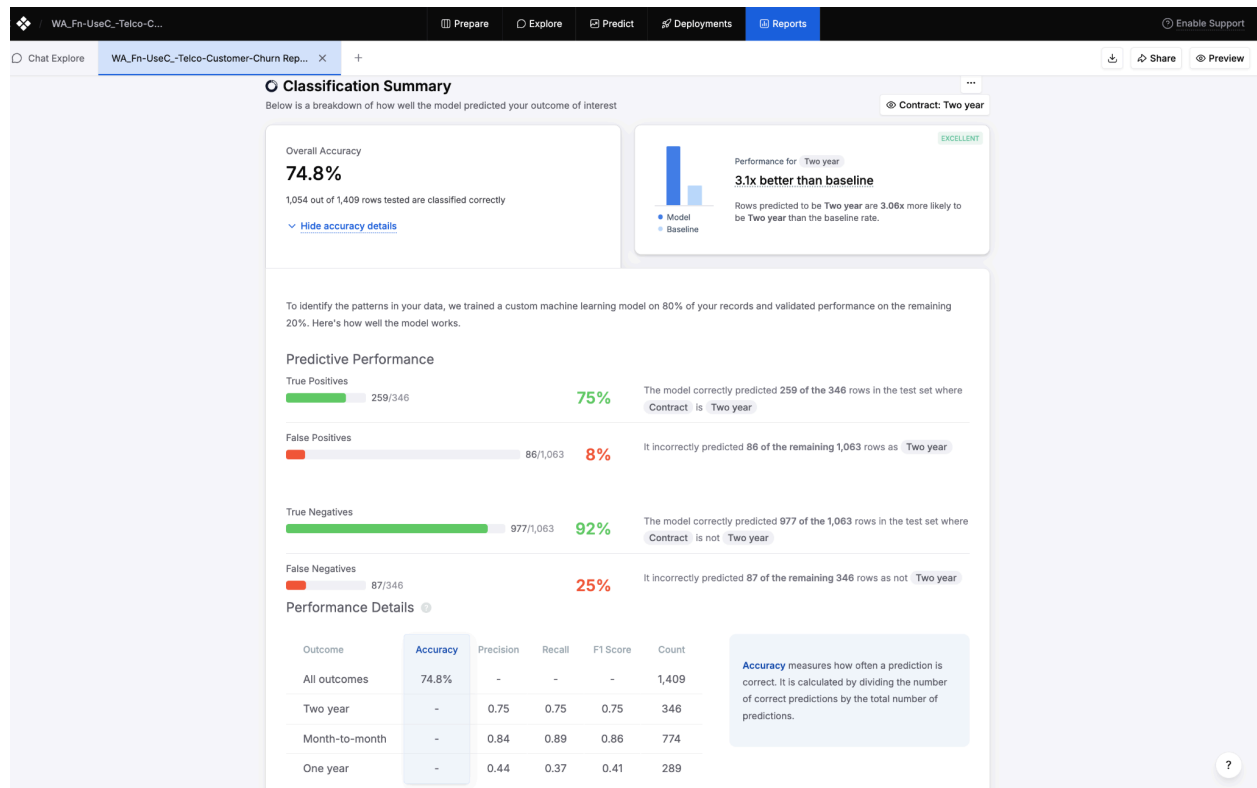
```
{
  "customerID": "",
  "gender": "",
  "SeniorCitizen": "0",
  "Partner": "",
  "Dependents": "No",
  "tenure": "1",
  "PhoneService": "",
  "MultipleLines": "",
  "InternetService": "DSL",
  "OnlineSecurity": "No",
  "OnlineBackup": "",
  "DeviceProtection": "No",
  "TechSupport": "",
  "StreamingTV": "",
  "StreamingMovies": "",
  "PaperlessBilling": "",
  "PaymentMethod": "",
  "MonthlyCharges": "",
  "TotalCharges": "",
  "Churn": ""
}
```

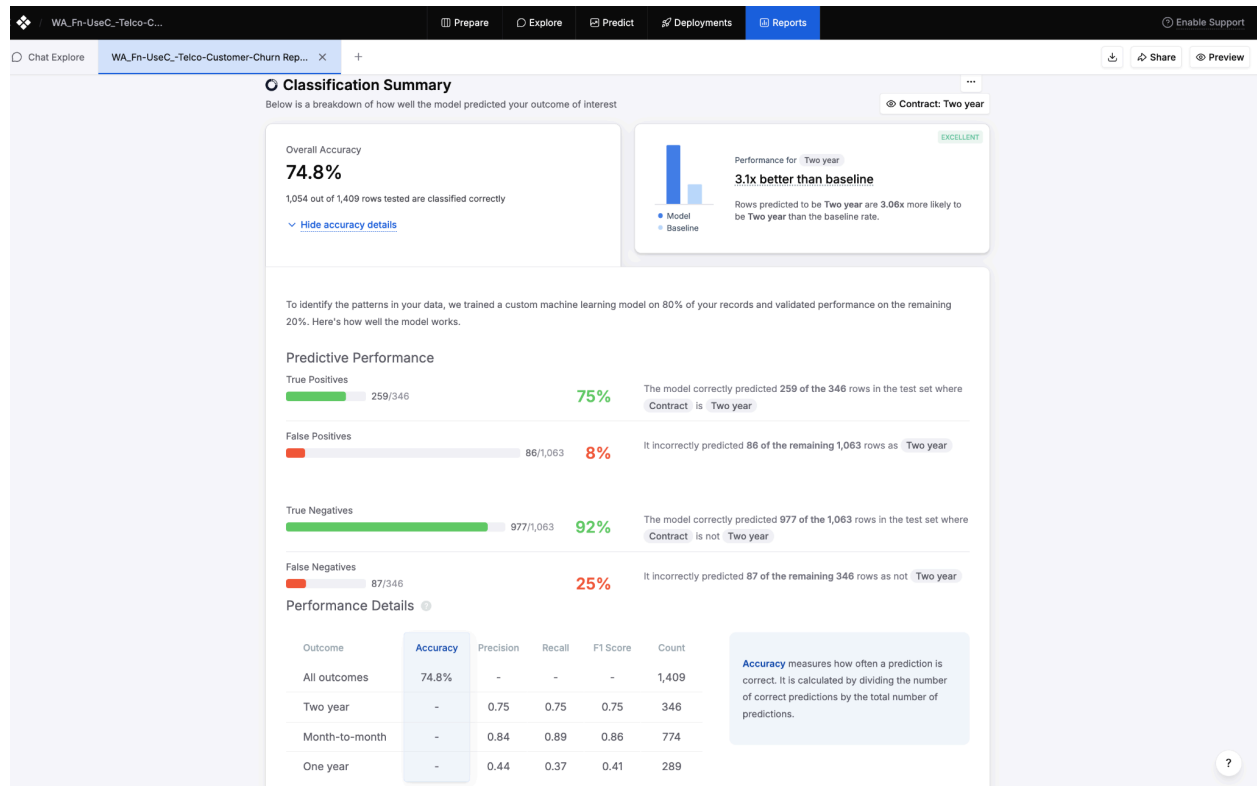
Step 5: Insights

Insights and Visualizations

After deployment, the predictions were analyzed to extract actionable insights:

- Customers with high monthly charges or short tenure were more likely to churn.
- Visualizations generated by Akkio highlighted these patterns, enabling stakeholders to make informed decisions about retention strategies.





5. Akkio.com provides:

1. User friendly interface
2. Zero Code (Totally UI based)
3. Data cleaning
4. AutoML
5. Model Evaluation
6. Prediction
7. Deployment
8. Visualization

Why Akkio?

Akkio stands out as a no-code AutoML platform due to its:

1. **User-Friendly Interface:** Simplifies complex ML workflows.
2. **Data Cleaning Automation:** Saves time by automating preprocessing steps.
3. **AutoML Features:** Selects the best algorithm and optimizes it for the given dataset.
4. **Deployment Options:** Easy integration into real-world applications.
5. **Comprehensive Insights:** Provides clear visualizations and interpretable results.

Conclusion

This assignment allowed me to experience the potential of no-code tools like Akkio in solving real-world business problems efficiently. The platform's ability to handle the entire machine learning lifecycle, from data preparation to deployment, underscores its value for both technical and non-technical users.